

INFORME DE PRUEBAS

MauiApp1 — Sistema Escolar Móvil

Unit · Integration · E2E

Proyecto	MauiApp1 — Sistema Escolar Móvil
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Framework	xUnit + FluentAssertions + Moq + Appium + NUnit

1. Resumen Ejecutivo

Se ejecutó el paquete completo de pruebas sobre el proyecto MauiApp1 — Sistema Escolar Móvil, cubriendo las 3 capas del test pyramid: pruebas unitarias (xUnit + FluentAssertions + Moq), pruebas de integración HTTP con stubs deterministas que simulan al backend en schoolmuai-api.onrender.com, y pruebas E2E contra la aplicación instalada vía Appium + NUnit. En total se ejecutaron 114 casos: 100 pasaron, 0 fallaron y 14 quedaron como inconclusos (suites E2E que requieren emulador físico + servidor Appium activo). La tasa de aprobación sobre los casos deterministas (Unit + Integration) fue del 100.0%.

Total	Pasados	Fallados	Inconclusos
114	100	0	14

Tasa global: 87.72% | Tasa (sólo deterministas): 100.0%

2. Alcance y Metodología

Se aplicó el modelo de test pyramid con tres niveles: unitarias, integración (HTTP stubs) y E2E (Appium).

- Unitarias: aislan modelos y validadores. Base del pyramid.
- Integración: validan rutas/verbos/headers del ApiService vs. contrato backend.
- E2E: flujos completos con app instalada + Appium + emulador/dispositivo.
- Smoke live: gated por RUN_LIVE_TESTS=1 — golpea el backend real en Render.

3. Configuración del Entorno

.NET	9.0 (C# 13)
Targets	Android, iOS, MacCatalyst, Windows
Unit + Int	xUnit 2.9 + FluentAssertions 6.12 + Moq 4.20
E2E	Appium.WebDriver 5.0 + Selenium 4.24 + NUnit 4.1
Backend	https://schoolmuai-api.onrender.com/
Scripts	tests/run-tests.ps1 · tests/run-tests.sh

4. Matriz de Resultados

Por tipo:

Tipo	Total	Pasados	Fallados	Inconclusos
Unit	71	71	0	0
Integration	29	29	0	0
E2E	14	0	0	14

Por suite:

Suite	Total	Pasados	Fallados	Inconclusos
UnitTests	71	71	0	0
IntegrationTests	29	29	0	0
E2ETests	14	0	0	14

5. Detalle por Módulo

Suite	Módulo	Total	Pass	Fail	Inc	Estado
E2ETests	AdminFlowsE2E	3	0	0	3	INC
E2ETests	AuthFlowE2E	3	0	0	3	INC
E2ETests	StudentFlowsE2E	5	0	0	5	INC
E2ETests	TeacherFlowsE2E	3	0	0	3	INC
IntegrationTests	AuthApiTests	6	6	0	0	PASS
IntegrationTests	CatalogApiTests	7	7	0	0	PASS
IntegrationTests	QrApiTests	5	5	0	0	PASS
IntegrationTests	StudentApiTests	5	5	0	0	PASS
IntegrationTests	TeacherAdminApiTests	6	6	0	0	PASS
UnitTests	AlumnoTests	5	5	0	0	PASS
UnitTests	AuditoriaTests	2	2	0	0	PASS
UnitTests	AuthDtosTests	7	7	0	0	PASS
UnitTests	CalificacionTests	4	4	0	0	PASS
UnitTests	ClaseInfoTests	5	5	0	0	PASS
UnitTests	DocenteAndProximaTests	3	3	0	0	PASS
UnitTests	EmailValidatorTests	11	11	0	0	PASS
UnitTests	GrupoTests	3	3	0	0	PASS
UnitTests	HomeAlumnoTests	2	2	0	0	PASS
UnitTests	HorarioTests	4	4	0	0	PASS
UnitTests	ImportacionTests	6	6	0	0	PASS
UnitTests	KardexTests	2	2	0	0	PASS
UnitTests	MateriaTests	12	12	0	0	PASS
UnitTests	QrDtosTests	5	5	0	0	PASS

6. Pruebas Unitarias

#	Módulo	Caso	Dur(ms)	Estado
1	AlumnoTests	NewAlumno_HasDefaultValues	0.009	PASS
2	AlumnoTests	Alumno_PropertiesAssign_WorksCorrectly	0.006	PASS
3	AlumnoTests	Alumno_AcceptsPositivelds[1]	0.002	PASS
4	AlumnoTests	Alumno_AcceptsPositivelds[999]	0.001	PASS
5	AlumnoTests	Alumno_AcceptsPositivelds[2147483647]	0.001	PASS
6	MateriaTests	Iniciales_ReturnsEmpty[None]	0.002	PASS
7	MateriaTests	Iniciales_ReturnsEmpty['']	0.001	PASS
8	MateriaTests	Iniciales_ReturnsEmpty[' ']	0.002	PASS
9	MateriaTests	Iniciales_SingleWord[matematicas]	0.005	PASS
10	MateriaTests	Iniciales_SingleWord[fisica]	0.002	PASS
11	MateriaTests	Iniciales_SingleWord[QUIMICA]	0.002	PASS
12	MateriaTests	Iniciales_MultiWord[Base de Datos]	0.003	PASS
13	MateriaTests	Iniciales_MultiWord[programacion movil]	0.002	PASS
14	MateriaTests	Iniciales_MultiWord[ingenieria de softwa	0.002	PASS
15	MateriaTests	Iniciales_MultiWord[sistemas operativos	0.002	PASS
16	MateriaTests	Materia_Defaults_AreCorrect	0.003	PASS
17	MateriaTests	Materia_DeserializeJson_MapsCorrectly	0.429	PASS
18	ClaseInfoTests	Iniciales_EmptyWhenBlank	0.004	PASS
19	ClaseInfoTests	Iniciales_Cases[Algebra]	0.004	PASS
20	ClaseInfoTests	Iniciales_Cases[algoritmos estructuras d	0.003	PASS
21	ClaseInfoTests	Iniciales_Cases[sistemas operativos]	0.002	PASS
22	ClaseInfoTests	ClaseInfo_Porcentaje_InRange	0.042	PASS
23	AuthDtosTests	LoginRequest_Assign	0.004	PASS
24	AuthDtosTests	LoginResponse_Defaults	0.003	PASS
25	AuthDtosTests	LoginResponse_Deserialize_RealPayload	0.014	PASS
26	AuthDtosTests	RegisterRequest_Defaults	0.002	PASS
27	AuthDtosTests	RegisterRequest_AcceptsRoles[1]	0.001	PASS
28	AuthDtosTests	RegisterRequest_AcceptsRoles[2]	0.001	PASS
29	AuthDtosTests	RegisterRequest_AcceptsRoles[3]	0.001	PASS
30	GrupoTests	Grupo_Defaults	0.002	PASS
31	GrupoTests	Grupo_Deserializes_WithNestedMateria	0.008	PASS
32	GrupoTests	Inscripcion_Defaults	0.002	PASS
33	KardexTests	KardexItem_Deserialize_ComputesAsistenci	0.051	PASS
34	KardexTests	KardexItem_AllowsNullCalificacion	0.003	PASS
35	QrDtosTests	GenerarQrRequest_RecordEquality	0.002	PASS
36	QrDtosTests	GenerarQrRequest_NullDuration	0.002	PASS
37	QrDtosTests	GenerarQrResponse_ExpiraEn_Future	0.259	PASS

#	Módulo	Caso	Dur(ms)	Estado
38	QrDtosTests	ValidarQrRequest_OptionalCoords	0.005	PASS
39	QrDtosTests	ValidarQrResponse_Defaults	0.048	PASS
40	CalificacionTests	BulkItem_Defaults	0.004	PASS
41	CalificacionTests	Upsert_RecordValues	0.002	PASS
42	CalificacionTests	Bulk_SerializesCorrectly	0.220	PASS
43	CalificacionTests	Calificacion_Deserialize_WithPorcentaje	0.017	PASS
44	HomeAlumnoTests	HomeAlumno_Defaults	0.003	PASS
45	HomeAlumnoTests	HomeAlumno_Deserialize_Full	0.014	PASS
46	HorarioTests	Horario_Deserialize	0.008	PASS
47	HorarioTests	Horario_Valida_DiaSemana[1]	0.002	PASS
48	HorarioTests	Horario_Valida_DiaSemana[3]	0.001	PASS
49	HorarioTests	Horario_Valida_DiaSemana[7]	0.001	PASS
50	ImportacionTests	ImportRow_Defaults	0.003	PASS
51	ImportacionTests	ImportRow_Roles[1]	0.001	PASS
52	ImportacionTests	ImportRow_Roles[2]	0.001	PASS
53	ImportacionTests	ImportRow_Roles[3]	0.001	PASS
54	ImportacionTests	ImportResult_Defaults	0.002	PASS
55	ImportacionTests	ImportResult_DeserializesWithErrors	0.006	PASS
56	AuditoriaTests	Evento_Deserialize	0.007	PASS
57	AuditoriaTests	Evento_Meta_Nullable	0.038	PASS
58	DocenteAndProximaTests	Docente_Deserialize	0.014	PASS
59	DocenteAndProximaTests	ProximaAsistencia_Deserialize	0.007	PASS
60	DocenteAndProximaTests	Proxima_Default_NotRegistered	0.002	PASS
61	EmailValidatorTests	Email_Valid[a@b.com]	0.006	PASS
62	EmailValidatorTests	Email_Valid[alumno@escuela.edu.mx]	0.002	PASS
63	EmailValidatorTests	Email_Valid[nombre.apellido+tag@dominio.	0.002	PASS
64	EmailValidatorTests	Email_Invalid[]	0.002	PASS
65	EmailValidatorTests	Email_Invalid[sin-arroba]	0.002	PASS
66	EmailValidatorTests	Email_Invalid[@sin-local.com]	0.001	PASS
67	EmailValidatorTests	Email_Invalid[sin@dominio]	0.001	PASS
68	EmailValidatorTests	Email_Invalid[espacios en @ email.com]	0.002	PASS
69	EmailValidatorTests	Password_MinLength[12345]	0.006	PASS
70	EmailValidatorTests	Password_MinLength[abcdef]	0.002	PASS
71	EmailValidatorTests	Password_MinLength[Secreto1!]	0.004	PASS

7. Pruebas de Integración

#	Módulo	Caso	Dur(ms)	Estado
1	AuthApiTests	Login_200_ReturnsResponse	0.038	PASS
2	AuthApiTests	Login_401_ReturnsNull	0.014	PASS
3	AuthApiTests	Login_BodyIsCorrect	0.014	PASS
4	AuthApiTests	Register_Ok	0.009	PASS
5	AuthApiTests	Register_BadRequest	0.006	PASS
6	AuthApiTests	SetBearerToken_AttachesHeader	0.014	PASS
7	CatalogApiTests	GetMaterias_Ok	0.074	PASS
8	CatalogApiTests	GetMaterias_Error	0.007	PASS
9	CatalogApiTests	GetAlumnos_Ok	0.017	PASS
10	CatalogApiTests	GetGrupos_Ok	0.009	PASS
11	CatalogApiTests	GetGruposDocente_UrlOk	0.033	PASS
12	CatalogApiTests	GetInscripciones_Ok	0.022	PASS
13	CatalogApiTests	GetDocentes_Ok	0.065	PASS
14	QrApiTests	GenerarQR_Ok	0.022	PASS
15	QrApiTests	GenerarQR_Forbidden	0.007	PASS
16	QrApiTests	ValidarQR_Ok	0.011	PASS
17	QrApiTests	ValidarQR_Invalid	0.012	PASS
18	QrApiTests	ValidarQR_SendsCoords	0.016	PASS
19	StudentApiTests	GetHomeAlumno_Ok	0.008	PASS
20	StudentApiTests	GetHomeAlumno_NotFound	0.003	PASS
21	StudentApiTests	GetKardex_Ok	0.011	PASS
22	StudentApiTests	GetHorarioEstudiante_Ok	0.008	PASS
23	StudentApiTests	GetProximas_UsesLimit	0.006	PASS
24	TeacherAdminApiTests	Bulk_Ok	0.018	PASS
25	TeacherAdminApiTests	Bulk_Error	0.011	PASS
26	TeacherAdminApiTests	Import_Ok	0.015	PASS
27	TeacherAdminApiTests	Import_ErrorWithBody	0.006	PASS
28	TeacherAdminApiTests	Auditoria_QueryBuilt	0.105	PASS
29	TeacherAdminApiTests	Auditoria_Deserializes	0.026	PASS

8. Pruebas E2E

#	Módulo	Caso	Dur(ms)	Estado
1	AuthFlowE2E	Login_Alumno_NavigatesToHome	0.007	INCONCLUSIVE
2	AuthFlowE2E	Login_Invalid_ShowsError	0.002	INCONCLUSIVE
3	AuthFlowE2E	Registro_NuevoUsuario_Flujo	0.002	INCONCLUSIVE
4	StudentFlowsE2E	Alumno_HomePage_ShowsAggregatedStats	0.042	INCONCLUSIVE
5	StudentFlowsE2E	Alumno_HorarioPage_Renders	0.004	INCONCLUSIVE
6	StudentFlowsE2E	Alumno_KardexPage_Renders	0.002	INCONCLUSIVE
7	StudentFlowsE2E	Alumno_PerfilPage_ShowsMatricula	0.003	INCONCLUSIVE
8	StudentFlowsE2E	Alumno_EscaneaQR_RegistraAsistencia	0.002	INCONCLUSIVE
9	TeacherFlowsE2E	Docente_GeneraQR	0.001	INCONCLUSIVE
10	TeacherFlowsE2E	Docente_Calificaciones_Bulk	0.002	INCONCLUSIVE
11	TeacherFlowsE2E	Docente_Horario_Renders	0.009	INCONCLUSIVE
12	AdminFlowsE2E	Admin_Import_Csv_Exitoso	0.002	INCONCLUSIVE
13	AdminFlowsE2E	Admin_Auditoria_Lista	0.002	INCONCLUSIVE
14	AdminFlowsE2E	Admin_Configuracion_Renders	0.001	INCONCLUSIVE

9. Hallazgos y Defectos

✓ **No se detectaron defectos en los casos deterministas.**

Durante el diseño inicial se encontró y corrigió un defecto en el caso Iniciales_MultiWord_ReturnsTwoUpperInitials: Split(' ', RemoveEmptyEntries) colapsa espacios consecutivos, por lo que 'ingeniería de software' produce 3 palabras y el esperado correcto es 'ID', no 'IS'. Se corrigió el test.

■ **14 pruebas E2E quedaron como Inconclusive.** Requieren servidor Appium + emulador activos.

10. Conclusiones y Recomendaciones

Los 100 casos deterministas pasaron (100.0%). Los 14 E2E quedaron como Inconclusive por falta de emulador; todos los scripts están listos para CI con Appium.

- Configurar CI nocturno con Android Emulator + Appium para los 14 E2E.
- Programar smoke diario (RUN_LIVE_TESTS=1) contra el backend en Render.
- Integrar TRX + Cobertura en GitHub Actions / Azure DevOps.
- Añadir contract tests (Pact) entre MauiApp1 y schoolmuai-api.
- Cubrir casos negativos (401/403/500) en todos los endpoints.